

Adaptive Spectrum-Aware Load Balancing for Multi-Access Edge Computing in Dense Vehicular Networks

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Abstract—The proliferation of connected and autonomous vehicles has placed unprecedented demand on multi-access edge computing (MEC) infrastructure deployed along dense urban corridors. Existing load-balancing schemes for vehicular MEC either assume static channel conditions or ignore the coupling between spectrum allocation and task-offloading latency, leading to congestion collapse at peak intersection density. This paper proposes a Spectrum-Aware Adaptive Load Balancer (SALB) that jointly optimizes edge-server task assignment and sub-channel allocation using a two-timescale Lyapunov drift-plus-penalty formulation. SALB decomposes the joint problem into a fast inner loop that solves per-slot channel assignment via a weighted bipartite matching relaxation, and a slow outer loop that adapts server admission thresholds based on a moving estimate of queue backlog. We evaluate SALB against three baselines – round-robin, greedy nearest-edge, and a deep reinforcement learning (DRL) scheme – on a simulated 5.8 GHz vehicular testbed spanning a 3.2 km urban corridor with 14 roadside units. SALB reduces mean task completion latency by 38.4% relative to greedy nearest-edge and by 21.7% relative to the DRL baseline at a vehicle density of 220 vehicles/km, while sustaining a 99.1% task admission rate under bursty arrival patterns. We further derive a closed-form bound on queue backlog under SALB and show empirically that the bound tracks simulated backlog within a 6% margin across all tested densities.

Index Terms—Multi-access edge computing, vehicular networks, load balancing, Lyapunov optimization, task offloading, spectrum allocation.

I. INTRODUCTION

THE shift toward cooperative perception and trajectory planning in connected vehicle fleets has moved a substantial share of onboard computation off the vehicle and onto roadside multi-access edge computing (MEC) nodes. Tasks such as cooperative object fusion, high-definition map alignment, and short-horizon trajectory re-planning are latency-critical: a completion deadline miss of even 40–60 ms can invalidate a maneuver decision at highway speed. As vehicle density increases at urban intersections and highway merge points, the roadside units (RSUs) that host MEC servers face two coupled bottlenecks – (i) contention for the limited number of orthogonal sub-channels available for task upload, and (ii) heterogeneous and time-varying compute queues at each edge server.

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Most prior work treats these two bottlenecks separately. Channel-allocation literature typically assumes a fixed, already-decided server assignment, while server load-balancing literature typically assumes an interference-free, infinite-capacity uplink. This separation is convenient for analysis but produces poor outcomes in practice: a load balancer that ignores channel contention may route many vehicles to a nominally under-loaded server that is, in fact, unreachable at usable rate because the serving RSU's sub-channels are saturated.

This paper makes the following contributions:

- 1) We formulate joint task-to-server assignment and sub-channel allocation for vehicular MEC as a two-timescale stochastic optimization problem and derive a Lyapunov drift-plus-penalty bound that admits a tractable per-slot decomposition.
- 2) We propose SALB, an online algorithm that solves the fast per-slot channel-assignment sub-problem via weighted bipartite matching and adapts slow-timescale admission thresholds from an exponentially weighted backlog estimate, requiring no offline training and no knowledge of future arrivals.
- 3) We validate SALB against round-robin, greedy nearest-edge, and a DRL baseline on a corridor-scale vehicular simulation, showing consistent latency and admission-rate improvements across a wide range of vehicle densities, and confirm the theoretical backlog bound empirically.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the system model. Section IV presents the SALB algorithm and its analysis. Section V reports experimental results, and Section VI concludes.

II. RELATED WORK

Task offloading in vehicular MEC has been studied extensively under the umbrella of vehicular fog and edge computing. Early work by Chen *et al.* [1] modeled offloading as a single-server queuing problem and derived stability conditions under Poisson arrivals, but did not consider multi-server contention. Subsequent work introduced multi-RSU cooperative offloading [2], using game-theoretic formulations to distribute load across neighboring servers; these approaches assume a shared, contention-free backhaul and do not model the wireless uplink explicitly.